

GSE98923 co-expression analysis

CEMiTool baseline - 54-sample balanced design

2026-09-19

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1 Scope

This report documents the computational analysis of the balanced 54-sample subset of GEO accession **GSE98923** used in the CEMiTool baseline workflow.

The report is intended as an analysis record and potential supplementary document for a scientific manuscript. Large machine-readable matrices are not printed in full in the document; complete files are preserved separately and indexed in the supplementary manifest.

2 Experimental design

The analysis contains **54 samples**, covering two cultivars, three developmental stages, three years, and three biological replicates per cultivar-stage-year combination.

Table 1: Sample counts by cultivar, developmental stage, and year.

Cultivar	Stage	Year	Freq
Cabernet Sauvignon	FruitSet	2012	3
Pinot noir	FruitSet	2012	3
Cabernet Sauvignon	Harvest	2012	3
Pinot noir	Harvest	2012	3

Cultivar	Stage	Year	Freq
Cabernet Sauvignon	Veraison	2012	3
Pinot noir	Veraison	2012	3
Cabernet Sauvignon	FruitSet	2013	3
Pinot noir	FruitSet	2013	3
Cabernet Sauvignon	Harvest	2013	3
Pinot noir	Harvest	2013	3
Cabernet Sauvignon	Veraison	2013	3
Pinot noir	Veraison	2013	3
Cabernet Sauvignon	FruitSet	2014	3
Pinot noir	FruitSet	2014	3
Cabernet Sauvignon	Harvest	2014	3
Pinot noir	Harvest	2014	3
Cabernet Sauvignon	Veraison	2014	3
Pinot noir	Veraison	2014	3

3 Expression data

The processed matrix used for CEMiTool contains **29971 genes** and **54 samples**.

Expression values were derived from the processed RPKM matrix distributed with GSE98923 and transformed as:

$$\log_2(\text{RPKM} + 1)$$

The complete matrices are available under:

- `data/processed/expression_rpkм.tsv`
- `data/processed/expression_log2rpkм.tsv`

4 CEMiTool analysis

CEMiTool identified **9 modules** in the baseline analysis.

The analysis used Pearson correlation, an unsigned network, a signed TOM, a minimum module size of 30, and a fixed random seed of 1234.

Automatic beta selection did not identify a usable value in the initial baseline run. To reproduce the established 54-sample baseline, the soft-thresholding power was fixed explicitly at **beta = 7** using `set_beta = 7` and `force_beta = FALSE`. This value is specific to the present expression matrix and should be re-evaluated if the sample set, normalization, or filtering changes. The recorded beta for this run is **7**.

Table 2: Recorded analysis parameters.

parameter	value
dataset	GSE98923
samples	54
transform	log2(RPKM+1)
correlation	pearson
network_type	unsigned
tom_type	signed
beta_strategy	manual_set_beta_baseline_reproduction
set_beta	7
force_beta	FALSE
beta	7
r2	0.549579273382905
filter	TRUE
filter_pval	0.1
min_module_size	30
seed	1234
modules	9

5 Diagnostic and analysis figures

All figures generated by the analysis are stored in **results/figures/**. PDF figures are converted to PNG copies for reliable inclusion in Word and HTML reports.

5.1 beta_r2

[1] “assets/figures/beta_r2.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.2 gsea

[1] “assets/figures/gsea.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.3 hist

[1] “assets/figures/hist.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.4 mean_k

[1] “assets/figures/mean_k.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.5 mean_var

[1] “assets/figures/mean_var.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.6 profile_M1

[1] “assets/figures/profile_M1.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.7 profile_M2

[1] “assets/figures/profile_M2.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.8 profile_M3

[1] “assets/figures/profile_M3.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.9 profile_M4

[1] “assets/figures/profile_M4.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.10 profile_M5

[1] “assets/figures/profile_M5.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.11 profile_M6

[1] “assets/figures/profile_M6.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.12 profile_M7

[1] “assets/figures/profile_M7.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.13 profile_M8

[1] “assets/figures/profile_M8.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.14 profile_Not_Correlated

[1] “assets/figures/profile_Not_Correlated.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

5.15 qq

[1] “assets/figures/qq.png” attr(“class”) [1] “knit_image_paths” “knit_asis”

6 Exported result tables

The following section provides a compact view of every tabular result exported by CEMiTool. Complete tables remain available as machine-readable files under `results/tables/`.

6.1 beta_fit_indices.tsv

Dimensions: 15 rows x 10 columns.

Table 3: beta_fit_indices.tsv

Power	SFT.R.sq	slope	truncated.R.sq	mean.k.	median.k.	max.k.	Density	Centralization	Heterogeneity
1	0.8394009	1.5707687	0.7946939	1561.857117	54.661221	111.98010	0.5122523	0.1805457	0.3207976
2	0.4232012	0.3873307	0.5948254	1055.584872	206.78781	705.72110	0.3462069	0.2133693	0.4790910
3	0.0123210	0.0444688	0.7254709	792.1069588	4.928451	453.85820	0.2597924	0.2171812	0.5801171
4	0.0875404	-	0.8310439	627.739736	75.701751	1274.27600	0.2058838	0.2121878	0.6544427
		0.1174076							
5	0.2744513	-	0.8129080	514.611285	28.426901	1136.65840	0.1687803	0.2041506	0.7137761
		0.2192085							
6	0.4333524	-	0.7362054	431.828944	21.508541	1026.44510	0.1416297	0.1951480	0.7635877
		0.3044837							
7	0.5495793	-	0.6942176	368.657523	39.871359	935.47990	0.1209110	0.1860264	0.8068294
		0.3627214							
8	0.6217077	-	0.6648029	318.950732	77.028728	58.72380	0.1046083	0.1771490	0.8452677
		0.4267637							
9	0.6650263	-	0.6414873	278.909912	28.173737	92.84740	0.0914759	0.1686699	0.8800415
		0.4781301							
10	0.7063742	-	0.6502824	246.049851	89.373377	35.54000	0.0806985	0.1606465	0.9119266
		0.5251829							
12	0.7067470	-	0.6230659	195.571181	34.480706	40.39660	0.0641427	0.1459880	0.9690884
		0.6147726							
14	0.7472853	-	0.6797739	158.919239	7.11593	564.34660	0.0521218	0.1330578	1.0196729
		0.6734742							
16	0.7496827	-	0.6990696	131.367807	2.38920	502.02300	0.0430855	0.1216459	1.0654044
		0.7344417							
18	0.7495778	-	0.7234977	110.099295	3.69994	449.96880	0.0361100	0.1115423	1.1073859
		0.7871557							
20	0.7605758	-	0.7500230	93.33028	40.81002	405.83510	0.0306101	0.1025615	1.1463678
		0.8268108							

6.2 enrichment_es.tsv

Dimensions: 8 rows x 7 columns.

Table 4: enrichment_es.tsv

pathway	CS_FruitSet	CS_Veraison	CS_Harvest	PN_FruitSet	PN_Veraison	PN_Harvest
M2	-0.3065545	-0.3989999	0.1714573	-0.1424825	0.4048774	0.3335254
M3	-0.4814519	-0.1082369	0.3755327	-0.4567550	0.2261734	0.5485116
M4	-0.6334693	-0.6877653	0.4837331	-0.6504614	-0.5596429	0.8691421
M5	-0.5119349	0.7918700	-0.2469925	-0.5727864	0.7598874	-0.2293153
M6	0.2911998	-0.6365587	0.4277549	0.2543111	-0.5997040	0.4927828
M7	-0.7218118	0.6143423	0.4761248	-0.7390699	0.7373351	0.5730595
M8	-0.4237900	-0.1594684	0.6348837	0.3063109	-0.6070465	0.5252492
Not.Correlated	-0.5779756	0.2377184	0.6996373	-0.5182987	-0.5736542	0.5720409

6.3 enrichment_nes.tsv

Dimensions: 8 rows x 7 columns.

Table 5: enrichment_nes.tsv

pathway	CS_FruitSet	CS_Veraison	CS_Harvest	PN_FruitSet	PN_Veraison	PN_Harvest
M2	-2.506928	-2.3022781	1.444438	-1.190401	1.8674508	2.580479
M3	-3.480997	-0.5548237	2.769954	-3.300488	0.9719954	3.715713
M4	-4.571961	-3.4839356	3.540738	-4.629760	-2.9350757	5.829017
M5	-3.593101	3.3696824	-1.356645	-4.032407	3.2201701	-1.312803
M6	1.371819	-3.0194682	2.885844	1.310250	-2.9118073	3.030700
M7	-4.148630	2.2601163	2.829898	-4.223805	2.6731864	3.099317
M8	-2.370546	-0.6591262	3.681302	1.404482	-2.5789567	2.825733
Not.Correlated	-2.455533	0.7075061	3.082636	-2.194883	-1.9110014	2.325400

6.4 enrichment_padj.tsv

Dimensions: 8 rows x 7 columns.

Table 6: enrichment_padj.tsv

pathway	CS_FruitSet	CS_Veraison	CS_Harvest	PN_FruitSet	PN_Veraison	PN_Harvest
M2	0.0000000	0.0e+00	0.0023112	0.1069297	0.0000006	0.0000000
M3	0.0000000	1.0e+00	0.0000000	0.0000000	0.5301775	0.0000000
M4	0.0000000	0.0e+00	0.0000000	0.0000000	0.0000000	0.0000000
M5	0.0000000	0.0e+00	0.0732519	0.0000000	0.0000000	0.0953020
M6	0.0707291	0.0e+00	0.0000000	0.1191895	0.0000000	0.0000000
M7	0.0000000	2.2e-06	0.0000005	0.0000000	0.0000000	0.0000000
M8	0.0000480	1.0e+00	0.0000000	0.1006289	0.0000003	0.0000004
Not.Correlated	0.0003294	1.0e+00	0.0000006	0.0015343	0.0045860	0.0007032

6.5 module.tsv

Dimensions: 3050 rows x 2 columns.

Table 7: module.tsv - first 40 rows shown

genes	modules
VIT_19s0014g02740	M1
VIT_08s0056g01610	M1
VIT_12s0059g00590	M1
VIT_01s0010g02030	M1
VIT_08s0056g01600	M1
VIT_05s0049g00520	M1
VIT_02s0025g03390	M1
VIT_11s0016g00210	M1
VIT_16s0022g00960	M1
VIT_07s0005g00660	M1
VIT_19s0090g01370	M1
VIT_06s0004g03240	M1
VIT_06s0004g04650	M1
VIT_05s0049g00760	M1
VIT_17s0000g03280	M5
VIT_19s0014g02730	M1
VIT_01s0011g05110	M7
VIT_03s0180g00070	M1
VIT_00s0187g00160	M1
VIT_18s0001g14270	M1
VIT_19s0090g01340	M1
VIT_09s0002g05180	M1
VIT_07s0031g02210	M1
VIT_19s0090g01000	M1
VIT_03s0038g01510	M1
VIT_16s0022g01770	M1
VIT_05s0094g00340	M1
VIT_18s0001g14260	M1
VIT_02s0025g01050	M1
VIT_19s0090g01350	M1
VIT_02s0033g01330	M1
VIT_02s0033g01120	M7
VIT_04s0008g05010	M1
VIT_02s0025g03600	M1
VIT_02s0025g04330	M1
VIT_13s0019g04280	M5
VIT_10s0003g00090	M1
VIT_03s0038g02520	M1
VIT_11s0016g04920	M1
VIT_12s0028g01060	M1

The complete table is provided as a supplementary machine-readable file.

6.6 parameters.tsv

Dimensions: 11 rows x 2 columns.

Table 8: parameters.tsv

Parameter	Value
filter	TRUE
apply_vst	FALSE
filter_pval	0.1
cor_method	pearson
min_ngen	30
merge_similar	TRUE
diss_thresh	0.8
r2	0.549579273382905
beta	7
phi	0.500166177070297
n_mods	9

6.7 selected_genes.txt

Dimensions: 3049 rows x 1 columns.

Table 9: selected_genes.txt - first 40 rows shown

VIT_19s0014g02740
VIT_08s0056g01610
VIT_12s0059g00590
VIT_01s0010g02030
VIT_08s0056g01600
VIT_05s0049g00520
VIT_02s0025g03390
VIT_11s0016g00210
VIT_16s0022g00960
VIT_07s0005g00660
VIT_19s0090g01370
VIT_06s0004g03240
VIT_06s0004g04650
VIT_05s0049g00760
VIT_17s0000g03280
VIT_19s0014g02730
VIT_01s0011g05110
VIT_03s0180g00070
VIT_00s0187g00160

VIT_19s0014g02740
VIT_18s0001g14270
VIT_19s0090g01340
VIT_09s0002g05180
VIT_07s0031g02210
VIT_19s0090g01000
VIT_03s0038g01510
VIT_16s0022g01770
VIT_05s0094g00340
VIT_18s0001g14260
VIT_02s0025g01050
VIT_19s0090g01350
VIT_02s0033g01330
VIT_02s0033g01120
VIT_04s0008g05010
VIT_02s0025g03600
VIT_02s0025g04330
VIT_13s0019g04280
VIT_10s0003g00090
VIT_03s0038g02520
VIT_11s0016g04920
VIT_12s0028g01060
VIT_18s0001g12830

The complete table is provided as a supplementary machine-readable file.

6.8 summary_eigengene.tsv

Dimensions: 9 rows x 55 columns.

Table 10: summary_eigengene.tsv

[illegible]

7 Supplementary data inventory

The full analysis files are indexed in `reports/supplementary/manifest.tsv`. The manifest records relative paths, file sizes, and MD5 checksums.

Table 13: Supplementary file inventory.

file	bytes	md5
data/metadata/phenotypes.tsv	1296	3ded458ab03ff3e1694c33db594c19f9
data/metadata/sample_geo_map.tsv	10654	c0e7260e2115a3fb85dd76228b953c44
data/metadata/sample_images.tsv	1497	2f2700c2e90786b44e078b0cfabdbfa7
data/metadata/samples.tsv	9957	494e1078fc9806589f3024eed79c8136
data/processed/expression_log2rpkm.rds	9656439	77f6b173e329b9c1583d1bd52c0ab5ca
data/processed/expression_log2rpkm.tsv	21802281	95761a42392a04d55642f378fe0fa5ce
data/processed/expression_rpkms.tsv	11158928	636a4103dc8b388862503c5c989b3690
logs/parameters.txt	300	3bf14dbb914cc65432f16277d255dbdf
logs/run_summary.txt	160	c57e247778a5e6ede7c1182847d3a3df
logs/session_info.txt	5832	46cfc2f7438b3a095a050a648622de80
reports/cemitoool/report.html	3273226	bd3e3c419ee2cedc094a607cd6594328
reports/diagnostics/diagnostics.html	975275	36fad49230c2f33891a2b863bf558472
reports/native_report_status.tsv	103	cb9c1034c7cf2e211fb29ce01b16b718
results/figures/beta_r2.pdf	5718	3a832fce3f39d8a42d7883008bff2485
results/figures/gsea.pdf	8691	9fbe6da8b9167c1b3ec36b265e3d951e
results/figures/hist.pdf	5105	72a13545484e062985873b38977282d0
results/figures/mean_k.pdf	5468	b359dd8954595a051756dad6d3f127ab
results/figures/mean_var.pdf	173388	ccd0ca62cf6f6162589074e8b45a5e5d
results/figures/profile.pdf	708378	81a6770fd5f7d522b70dd0f9008ae6e6
results/figures/qq.pdf	1392120	711b124828b3f13ffb558df365003e91
results/figures/sample_tree.pdf	6798	b1435209413630ef2bf502e4ec79e086
results/tables/beta_fit_indices.tsv	2527	6ec616e12b6c75d209918578a4a717cb
results/tables/enrichment_es.tsv	1036	1c414bc4170896a4d513cc8e0d49e66e
results/tables/enrichment_nes.tsv	996	afdf181bbe6c681be26c68a09bd6b5a9
results/tables/enrichment_adj.tsv	1080	cc9765eeca9374800fdeea198a80ee56
results/tables/module.tsv	79523	468f068ba331a8b00ee6f4a01f1380fb
results/tables/modules_genes.gmt	70337	2aa3050a38c637b22eefaae52b17ec68
results/tables/parameters.tsv	245	0566c86fb8add878d1102c1d450f1f36
results/tables/selected_genes.txt	57950	3fbe5f3efd8f2a6f69851ea0df1e0ad7
results/tables/summary_eigengene.tsv	10036	ae0d3e6048c1f931e59f144c7bad95ab
results/tables/summary_mean.tsv	9085	7f72cf2361ec104a219485674f02bb9e
results/tables/summary_median.tsv	7945	d7aa659158364cd3032e7fda7a76ac58

8 Reproducibility

```
## ```text
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
```

```

## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Spanish_Chile.utf8  LC_CTYPE=Spanish_Chile.utf8    LC_MONETARY=Spanish_Chile
## [4] LC_NUMERIC=C                    LC_TIME=Spanish_Chile.utf8
##
## time zone: America/Santiago
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] CEMiTool_1.36.0      GEOquery_2.80.0      Biobase_2.72.0      BiocGenerics_0.58.1 generic
##
## loaded via a namespace (and not attached):
##   [1] splines_4.6.1          ggplotify_0.1.3      R.oo_1.27.1          t
##   [5] ggpp_0.6.1             polyclip_1.10-7      preprocessCore_1.74.0 er
##   [9] confintr_1.0.2         XML_3.99-0.24        rpart_4.1.27         1
##  [13] httr2_1.3.0            fastcluster_1.3.0    doParallel_1.0.17    pr
##  [17] lattice_0.22-9         MASS_7.3-65          backports_1.5.1      ma
##  [21] limma_3.68.5           Hmisc_5.3-0          rmarkdown_2.32       ya
##  [25] otl_0.2.0              ggtangle_0.1.2        zip_3.0.2            co
##  [29] DBI_1.3.0              RColorBrewer_1.1-3   abind_1.4-8          G
##  [33] R.utils_2.13.0         purrr_1.2.2          pracma_2.4.6         y
##  [37] nnet_7.3-20            tweenr_2.0.3          rappdirs_0.3.4       a
##  [41] gdttools_0.5.1         IRanges_2.46.0       S4Vectors_0.50.3     er
##  [45] ggrepel_0.9.8          tidytree_0.4.8       ggpmisc_1.0.0        re
##  [49] MatrixModels_0.5-4     codetools_0.2-20     DelayedArray_0.38.2  ge
##  [53] DOSE_4.5.0             DT_0.34.0            xml2_1.6.0           t
##  [57] aplot_0.3.1            farver_2.1.2          matrixStats_1.5.0    st
##  [61] dynamicTreeCut_1.63-1  base64enc_0.1-6       Seqinfo_1.2.0        j
##  [65] Formula_1.2-6          survival_3.8-6        iterators_1.0.14     sy
##  [69] foreach_1.5.2          tools_4.6.1          ggnewscale_0.5.2     tr
##  [73] sna_2.8                Rcpp_1.1.2           glue_1.8.1           g
##  [77] SparseArray_1.12.2     mgcv_1.9-4           xfun_0.60            q
##  [81] MatrixGenerics_1.24.0  ggthemes_6.0.0        dplyr_1.2.1          w
##  [85] BiocManager_1.30.27    fastmap_1.2.0         SparseM_1.84-2        ca
##  [89] digest_0.6.39          R6_2.6.1             gridGraphics_0.5-1   co
##  [93] GO.db_3.23.1           RSQLite_3.53.3        R.methodsS3_1.8.2     t
##  [97] fontLiberation_0.1.0   data.table_1.18.6.1   httr_1.4.9           ht
## [101] S4Arrays_1.12.0        scatterpie_0.2.6      pkgconfig_2.0.3      g
## [105] blob_1.3.0             S7_0.2.2             impute_1.86.0        X
## [109] clusterProfiler_4.20.0 htmtltools_0.5.9      fontBitstreamVera_0.1.1 f
## [113] scales_1.4.0           png_0.1-9            ggfun_0.2.1          g

```

```

## [117] knitr_1.52          rstudioapi_0.19.0      tzdb_0.5.0
## [121] curl_8.0.0           coda_0.19-4.1          statnet.common_4.13.0
## [125] nlme_3.1-169         cachem_1.1.0           stringr_1.6.0
## [129] foreign_0.8-91       AnnotationDbi_1.74.0   pillar_1.11.1
## [133] vctrs_0.7.3          tidydr_0.0.6           cluster_2.1.8.2
## [137] evaluate_1.0.5       tinytex_0.61           readr_2.2.0
## [141] compiler_4.6.1       rlang_1.3.0            crayon_1.5.3
## [145] labeling_0.4.3       plyr_1.8.9             fs_2.1.0
## [149] stringi_1.8.9        network_1.20.0         WGCNA_1.74
## [153] Biostrings_2.80.2    lazyeval_0.2.3         fontquiver_0.2.1
## [157] quantreg_6.1         Matrix_1.7-5           hms_1.1.4
## [161] bit64_4.8.6          ggplot2_4.0.3          KEGGREST_1.52.2
## [165] SummarizedExperiment_1.42.0 igraph_2.3.3           memoise_2.0.1
## [169] fastmatch_1.1-8      bit_4.6.0              gson_0.2.1
## [173] polynom_1.4-1
##
## ```

```

9 Notes for manuscript use

This document is a computational analysis report, not a substitute for biological interpretation. Statements about biological mechanism, causal relationships, novelty, and comparison with previous literature should be supported separately with appropriate experimental evidence and verified references.